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MILITARY SPECIFICATION

DETECTING SET, MINE; PORTABLE; METALLIC AND NONMETALLIC; AN/PRS-7

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a portable detecting set for locating buried metallic and nonmetallic mines.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

NN-P-71	- Pallets, Material Handling Wood Stringer Construction 2-Way and 4-Way (Partial).
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6665

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MILITARY

MIL-B-18	- Battery, Dry.
MIL-I-46058	- Insulating Compound, Electrical (For Coating Printed Circuit Assemblies).

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ME

TA13217E1450	- Detecting Set, Mine, Portable, Metallic and Nonmetallic.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.4 and 6.3). Any changes or deviations of detecting sets from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish detecting sets conforming to this specification.

3.2 Design and construction. The detecting set shall be in accordance with Top Assembly, TA13217E1450 and as specified herein. The detecting set shall consist of a mine detector, a carrying strap, a carrying case, a replacement module, and an arctic carrying case for the battery. Unless otherwise specified (see 6.2), two batteries conforming to MIL-B-18 and drawing 13217E1498, shall be furnished by the Government for inclusion with each detecting set prior to shipment by the contractor. The mine detector shall consist of a search head, handle, and control box with headset.

3.2.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be available upon request for inspection by the contracting officer or his designated representative.

3.3 Materials. Materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Materiel deterioration and control. The detecting set shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the item may be exposed.

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3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3. Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the detecting sets may be newly fabricated from recovered materials to the maximum extent practicable, provided the detecting sets produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the detecting sets.

3.4 Parts. All parts used in fabrication shall be free of defects and imperfections that might affect safety, performance or serviceability of the finished product.

3.5 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with an insulating compound conforming to MIL-I-46058.

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the insulating compound shall not be treated.

When used, the insulating compound shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the insulating compound.

3.6 Performance requirements. The performance requirements of this specification apply over the complete range of environmental and battery voltage conditions; with the exception that the battery shall not be required to operate below +40° F. The performance requirements stated in 3.6.1 to 3.6.11 of this specification shall be met when the search head of the mine detector encounters the target from both the left and right side. These measurements shall be performed using a field simulator (see 4.6.2 and 6.4.2).

3.6.1 Electrical balance. With the search head positioned 1 inch from the target and tested as specified in 4.6.2.1 the output voltage across the headset of the mine detector shall be as follows:

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a. Less than 15 MVAC rms in search position.
b. Less than 45 MVAC rms in point high position.
c. Less than 12 MVAC rms in point mid position.
d. Less than 4 MVAC rms in point low position.

3.6.1.1 Balance at 2 inches. With the search head positioned 2 inches from the target and tested as specified in 4.6.2.1, the output voltage across the headset of the mine detector shall be as specified in 3.6.1.

3.6.1.2 Handle and cable interference. The handle assembly or cable assembly shall not affect the output null when moved as would be experienced in field conditions.

3.6.2 Detection. With the search head positioned 1 inch from the target and tested as specified in 4.6.2.1, the output voltage across the headset shall be as follows:

a. Between 1070 MVPP and 1735 MVPP in search position.
b. Between 78 MVAC and 315 MVAC rms in point high position.
c. Between 16 MVAC and 85 MVAC rms in point mid position.
d. Between 5 MVAC and 25 MVAC rms in point low position.

3.6.2.1 Detection at 2 inches. With the search head positioned 2 inches from the target and tested as specified in 4.6.2.1, the output voltage across the headset shall be as specified in 3.6.2.

3.6.3 Overload recovery time. With the detector selector in the search position, the voltage across the headset shall return to electrical balance (see 3.6.1) within 0.5 seconds after detection (see 3.6.2).

3.6.4 False signal rate. In the search position, and with sweep rate between 0.1 and 0.3 feet per second, the voltage across the headset shall be down at least 6 decibels (dB) from the detection voltage (see 3.6.2) when tested in accordance with 4.6.2.1.

3.6.5 Low voltage aural monitor. The mine detector shall indicate when either positive or negative battery voltage has decreased to 10.0 $\pm$0.6 volts, dc. The output voltage across headset shall have an amplitude greater than 1.0 volts peak to peak and a fundamental frequency between 1 and 10 pulses per second, when operated in the point high mode.

3.6.6 Current drain. With the mine detector selector switch in the search position and the detector in a null position (see 3.6.2) the current drain shall be as follows:

a. Positive supply - less than 20 milliamperes at voltage of 14.5 $\pm$0.5 volts, dc.
b. Negative supply - less than 60 milliamperes at voltage of 14.5 $\pm$0.5 volts, dc.

3.6.7 Battery voltage. The mine detector shall meet all performance requirements specified herein for all battery voltages down to 9.4 volts at

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which the detector still operates prior to cessation of operation due to the low voltage monitor, which will never be below 4.8 volts.

3.6.8 Stability. The mine detector when tested in accordance with 4.6.2.1 and retested twice in accordance with 4.6.2.1 at time intervals of not less than 2 days shall meet the requirements of 3.6.1 through 3.6.7.

3.6.9 Immersion. The mine detector, except headset assembly, shall exhibit no leakage and shall not be damaged when tested as specified in 4.6.2.4.

3.6.10 Vibration. The mine detector shall not be damaged when tested as specified in 4.6.2.2.

3.6.11 Bounce. The detecting set shall not be damaged when tested as specified in 4.6.2.3.

3.7 Temperature, altitude, and humidity.

3.7.1 Operation. The mine detector shall suffer no loss of performance while operating under temperature shocks within the range of +50° C to -30° C at elevations from sea level to 14,000 feet.

3.7.2 Storage. The detecting set shall not be damaged by storage under the following conditions:

a. Temperature shocks within the range of +70° C to -55° C.
b. Ambient temperature from +70° C to -55° C at all relative humidities possible at these temperatures.
c. Elevations from sea level to 40,000 feet and temperatures to -55° C, as encountered during transportation by air.

3.8 Reliability. The mine detector shall demonstrate a specified Mean-Time-Between-Failure (MTBF) of 1000 hours and the lower test MTBF shall be 500 hours when tested in accordance with 4.6.2.9.

3.9 Treatment and painting. The portions of the detecting set normally painted shall be cleaned, treated, and painted, requirements as specified on the drawings.

3.10 Identification marking. The detecting set shall be identified where specified herein in accordance with MIL-STD-130. The marking shall be applied to plates as specified on the drawings. The plates shall be attached as specified on the drawings.

3.11 Safety. The equipment shall be designed to eliminate unnecessary sharp corners or edges, excessive length of fastening devices and shall meet applicable criteria of MIL-STD-454, requirement 1.

3.12 Government-loaned property. The following property in the quantities indicated will be loaned by the Government (see 6.6).

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Item No.	Description	Quantity
1	Mine Detector Field Simulator	1

3.13 Government furnished property. The following property will be furnished by the Government for use in performing tests (see 6.7).

Item No.	Description	Identification
1	Battery, Dry, BA-1844/U	6135-00-461-3590

3.14 Workmanship. The workmanship in the detecting set shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The Government shall approve all test procedures and test standards that result from this specification.

4.1.1 Disassembly inspection. Failure of any test by the first article shall be cause for disassembly, in the presence of a Government representative, of the first article unit to the extent necessary to determine the cause of the failure. Each disassembled part shall be inspected in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

4.1.2 Parts and components inspection. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4.3 and 6.5).

4.1.3 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).
c. Inspection comparison (see 4.7).
d. Inspection of packaging (see 4.8).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in applicable paragraphs in the specification.

4.4 First article inspection. First article inspection shall be applied to the detecting sets as specified in 3.1. First article inspection shall consist of the examinations as specified in 4.6.1 and all tests indicated by "X" in column 2 of table I. Presence of one or more defects shall be cause for rejection. Failure of any test shall be cause for performing the disassembly inspection specified in 4.1.1.

4.4.1 Inspection failure. Failure of a first article detecting set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the first article detecting set(s) until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for tests shall be conducted in accordance with MIL-STD-105, level II. AQL shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects. Lots shall be formed from detecting sets which have successfully completed the examination and individual tests specified in 4.5.3.1.

4.5.2 Examination. Each detecting set shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection, and shall be cause for refusal by the Government to continue acceptance of production detecting sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on detecting sets previously accepted and produced under the contract.

4.5.3 Tests.

4.5.3.1 Individual. Each mine detector shall be subjected to the field simulation indicated by "X" in column 3 of table I. Failure of any test shall be cause for rejection.

4.5.3.2 Samples. Samples selected in accordance with 4.5.1 shall be subjected to the tests indicated by "X" in column 4 of table I. No adjustments

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shall be made on the detecting set after completion of the individual tests. AQL shall be 1.0 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The detecting sets shall be examined as specified herein for the following defects:

	Major	Requirement Paragraph
101.	Omission of any part or component.	3.2
102.	Assembly not as specified.	3.2
103.	Materials or components not as specified.	3.3
104.	Dimensions that affect form, fit, or function.	3.2.1
105.	Any part not in accordance with the QAP requirements shown on the drawings.	3.2.1
106.	Safety requirement not as specified.	3.11
107.	Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.	3.3.1
108.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.2
109.	Contractor does not have documentation available for identification of material, material finishes, or treatments.	3.3.3.
	Minor	
201.	Identification markings not as specified.	3.10
202.	Workmanship not as specified.	3.14
203.	Treatment and painting not as specified.	3.9
204.	Dimensions that do not affect form, fit, or function.	3.2.1
205.	Any drawing requirement other than QAP requirements not as shown on drawings.	3.2.1

4.6.2 Tests. The field simulation test in 4.6.2.1 shall be conducted using a field simulator (see 6.4.2) consisting of console, marble block, motor, motor control panel, carriage, and documentation; and an R-29 Test Set with documentation and test procedures. The field simulator and R-29 Test Set will be loaned by the Government (see 6.6). Where "damage" is referred to herein, it shall include, but not be limited to the following:

a. Breakage, permanent deformation, loosening, or failure of any part of a component.
b. Deterioration or corrosion of any hardware or finish.
c. Failure to operate in accordance with the requirement of this specification.

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TABLE I. Test schedule.

Test Sequence	First Article	Quality Conformance		Test	Test Para.	Requirement Paragraph
		Indiv.	Sample			
1	2	3	4	5	6	7
7	x	x	-	Field simulation.	4.6.2.1	3.6.1 thru 3.6.7
5	x	-	x	Vibration.	4.6.2.2	3.6.10
4	x	-	-	Bounce	4.6.2.3	3.6.11
6	x	-	x	Immersion.	4.6.2.4	3.6.9
3	x	-	x	Temperature operation and thermal shock.	4.6.2.5	3.7.1
2	x	-	-	Altitude and temperature storage.	4.6.2.6	3.7.2
1	x	-	-	High temperature and humidity storage.	4.6.2.7	3.7.2
*	x	-	x	Stability.	4.6.2.8	3.6.8
*	x	-	-	Reliability.	4.6.2.9	3.8

* No specific order is suggested.

4.6.2.1 Field simulation. Each mine detector shall be placed in a field simulator and tested for compliance with requirements of 3.6.1 through 3.6.7. For each mine detector, this test shall be conducted for the following operations: (1) The left side of the search head initially encounters the target (see 6.4.3) and (2) The right side of the search head encounters the target (see 6.4.3). Noncompliance with any requirement shall constitute failure of this test.

4.6.2.2 Vibration resistance. Test in accordance with MIL-STD-810, method 514.3, test procedure I, test condition I-3.2.12 using figure 5.4.3-36 for random vibration, vibrate for 15 minutes on each axis. After vibration, examine the mine detector for damage and then test in accordance with 4.6.2.1. Any damage to the mine detector or failure to operate as specified shall constitute failure of this test.

4.6.2.3 Bounce. Test the detecting set on a package testing machine capable of maintaining a constant speed of 260 rpm (4-1/3 cycles per second) with an amplitude of 1/2-inch (1 inch total excursion). Place the detecting set between the "fences" with 1-inch spare or "throw". Bounce the detecting set in the three mutually perpendicular directions; that is, the ends, the sides, and the top and bottom of the detecting set case shall be subjected to the bumping, each for a period of 15 minutes (total time 45 minutes). Examine the detecting set for damage; test in accordance with 4.6.2.1. Any damage to the detecting set or failure to operate as specified shall constitute failure of this test.

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4.6.2.4 Immersion. Test the mine detector, except headset, in accordance with MIL-STD-810, method 512.1, procedure III. The search head assembly and control box assembly, except headset, shall be completely immersed in water such that the uppermost parts of the assemblies are 2 ±1 inches below the surface of the water. The temperature of the water shall be 25° ±15° C. The absolute pressure of the air above the water shall be reduced to 1 inch of mercury (absolute) and then increased to 2.5 inches and maintained for 15 minutes. Bubbles emanating from within the mine detector shall be considered a leakage; however, bubbles which result from trapped air on the various exterior surfaces of the mine detector shall not be considered a leak. If a leakage occurs, the test shall be discontinued and the detector shall be removed from the tank. Examine the mine detector for damage; test in accordance with 4.6.2.1 within 15 minutes after removal from water. Any leakage or damage to the mine detector or failure to pass the test specified in 4.6.2.1 shall consitute failure of this test.

4.6.2.5 Temperature operation and thermal shock. Place the mine detector in the temperature chamber and reduce the temperature to -30° C ±2° C. Maintain this temperature for 3 hours. The mine detector shall then be removed from the chamber and within 15 minutes tested to meet the requirements of 3.6.1 and 3.6.2. Duplicate for six continuous test cycles. Nonconformance to 3.6.1 or 3.6.2 shall constitute failure of the low-temperature operation test. Place the mine detector in the chamber and increase the temperature to +50° C ±2° C. Maintain this temperature for 3 hours, then remove the mine detector from the chamber. Test to meet the requirements of 3.6.1 and 3.6.2, starting within 10 minutes and completing within 30 minutes after removal from the chamber. Duplicate for six continuous test cycles. Nonconformance to 3.6.1 or 3.6.2 shall constitute failure of the high-temperature operation test. For conformance sample testing under this paragraph, only one cold and one hot test cycle is required.

4.6.2.6 Low temperature and altitude storage. Place the detecting set in the altitude test chamber. Reduce the internal absolute pressure to 5.5 inches of mercury (corresponding to an altitude of 40,000 feet above sea level) and reduce the temperature to -55° C ±2° C. Maintain this pressure and temperature for 72 hours. The pressure in the chamber shall then be increased at a rate of not less than 3.0 inches of mercury per minute until atmospheric pressure has been reached. Remove the mine detector from its carrying case, and, at least 1 hour later, examine for damage and test in accordance with 4.6.2.1. Any damage to the mine detector or failure of 4.6.2.1 shall constitute failure of this test.

4.6.2.7 High-temperature and humidity storage. Test the detecting set (stored in its carrying case) in accordance with MIL-STD-202, method 106, except the upper temperature limit shall be +70° C ±2° C and step 7 shall not be performed. Polarization and loading voltages are not applicable. Upon completion of step 6 (see MIL-STD-202, method 106) of the final cycle, the detecting set shall be removed from the chamber to an ambient temperature of 25° C ±5°C. Remove the mine detector from its carrying case, examine for damage, and test in accordance with 4.6.2.1. Any damage to the mine detector or failure of 4.6.2.1 shall constitute failure of this test.

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4.6.2.8 Stability. This test shall consist of testing the same mine detector that successfully passed the test in 4.6.2.1 two more times in accordance with 4.6.2.1 at time intervals of not less than 2 days between any two tests. Nonconformance to 3.6.8 shall constitute failure of this test.

4.6.2.9 Reliability. Test the mine detector in accordance with figure 1 and MIL-STD-781, test plan XXC using the lower test MTBF. Mine detectors tested for reliability purposes shall not have more than 620 hours of test time nor less than 205 hours of test time. An external power supply may be used. Test the mine detector after every 5th cycle of reliabilty testing in accordance with 4.6.2.1 and at completion of the reliability test. For the purpose of determining the MTBF, a reliability failure is defined as any malfunction of the detector which causes or may cause failure to commence operation, cessation of operation, or degradation of performance outside specified limits; serious damage to the detector by continued operation or serious personnel safety hazards. Nonconformance to 3.8 or inability to pass the test in 4.6.2.1 shall constitute failure of this test.

4.6.2.10 Safety. The mine detector shall be examined initially and observed throughout testing to determine compliance with 3.11. Nonconformance shall constitute failure of the safety requirement.

4.7 Inspection comparison. The Government may select detecting sets at any time during the contract production period and subject these detecting sets to the examination and the tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article detecting sets. In addition to any test specified as part of the inspection comparison the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison detecting set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison detecting set(s) and shall be cause for refusal by the Government to continue acceptance of production detecting sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on detecting sets previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all detecting sets accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

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4.8.1.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, level II.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

Major

110. Case not closed, secured, and tamperproof seals attached as specified.
111. Box not as specified for level A.
112. Pallets not as specified for level A.
113. Load arrangement not as specified for level A.
114. Height greater than specified for level A.
115. Strapping not as specified for level A.
116. Battens not positioned and secured as specified for level A.
117. Packing not in accordance with the referenced document for commercial.
118. Marking missing, illegible, incorrect, or incomplete for level A or commercial.

5. PACKAGING

5.1 Preservation. The components comprising a complete detecting set shall be placed within the case. The case shall be closed and secured by the attached clasps. A minimum of two tamperproof seals shall be attached to the clasps in a manner that the case cannot be opened without destroying the seals (see 6.8).

5.2 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Each detecting set shall be individually packed in a close-fitting box conforming to PPP-B-636, grade V3c. Closure shall be in accordance with the appendix to the box specification. Unless otherwise specified (see 6.2), the packed detecting sets shall be palletized. The pallets shall conform to NN-P-71, type IV. Load arrangement shall conform to load pattern 3, Appendix D, of MIL-STD-147. The height of the pallet including the load shall not exceed 43 inches. The detecting sets shall be secured to the pallet with a total of five primary and secondary straps. Straps shall conform to QQ-S-781, type I or IV, finish A. Wood battens of not less than 1/2-inch thickness and not less than 2 inches wide shall be positioned under the vertical portions of the straps and straps held in place by staples.

5.2.2 Commercial. Detecting sets shall be packed in accordance with ASTM D 3951.

5.3 Marking (6.8).

5.3.1 Military. Marking for military levels of protection (level A) shall be in accordance with MIL-STD-129.

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5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951. In addition, weight and cube data shall be marked on each packed detector, and on each palletized unit load if applicable.

6. NOTES

6.1 Intended use. This specification covers a detecting set intended for use over all types of soil for the location of buried metallic and nonmetallic mines.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. Whether batteries will not be furnished (see 3.2).
d. Time frame required for submission of first article and number of detecting sets to be furnished (see 3.1).
e. Whether the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests shall be conducted by the contractor (see 3.1).
f. Degree of packing required (see 5.2).
g. Whether palletization is not required (see 5.2.1).

6.3 First article. When a first article inspection is required, the items should be detecting sets produced by production tooling. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the document's first article.

6.4 Definitions.

6.4.1 Module. A module consists of plug-in throw-away electronic assembly.

6.4.2 Field simulator. A test fixture, consisting of a field simulator and associated electronic equipment, used to simulate detection of targets in a minefield.

6.4.2.1 Description. The field simulator should consist of a standard target and standard background. The background media should be large in terms of wavelengths (greater than 1 at 300 MHz in all directions). The dielectric constant of this media should be consistent so that it will not change with time or environments. The target should be implanted in this background media and should have a consistent dielectric constant. The ratio of the dielectric constant of the target when compared to the background should be approximately 1 to 4. The standard target should be of a size to simulate the electrical signal specified in 3.6.2.

6.4.2.2 Stability. The field simulator shall be electrically stable in order to repeat test data and requirements on any given item from day to day without any external interference.

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6.4.2.3 Simulation. The field simulator shall simulate field conditions as close as possible in order to relate test results to actual detection operations.

6.4.2.4 Controls. The field simulator shall be such that a search head of mine detector can be fixed in relation to the target location. This location shall be repeatable on any unit at any time.

6.4.2.5 Performance requirements. The field simulator shall be capable of testing at 2-inch height position and at a speed of about 1.5 feet per second, and shall be used to test the following requirements of this specification.

a. Electrical balance (3.6.1).
b. Detection (3.6.2).
c. Overload recovery time (3.6.3).
d. False signal rate (3.6.4).
e. Low voltage aural monitor (3.6.5).
f. Current drain (3.6.6).
g. Battery voltage (3.6.7).
h. Stability (3.6.8).

6.4.2.6 Repeatability. The field simulator shall be capable of performing all the above tests as required. The results of these tests shall be repeatable at any given time and comparable to field conditions.

6.4.3 Definition of QAP. A QAP is a contractual requirement that indicates the minimum requirements which must be inspected on the product drawings to verify design objectives of the product and assure interchangeability of repair parts.

6.5 Quality assurance provisions (QAP). The contracting officer shall require the contractor to maintain records of all QAP inspections. The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrective action taken when applicable.

6.6 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.12.

6.7 Government furnished property. The contracting officer will arrange to furnish property specified in 3.13.

6.8 Preservation. Degrees of preservation have not been included as the preservation specified is the minimum acceptable for any level of protection. For purposes of preservation/packing level marking, the preservation shall be considered level A.

6.9 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, reports, and contractor's maintenance and operation manual to be furnished with each detecting set.

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6.10 Provisioning. The contracting officer should include provisioning requirements for the repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of detecting sets.

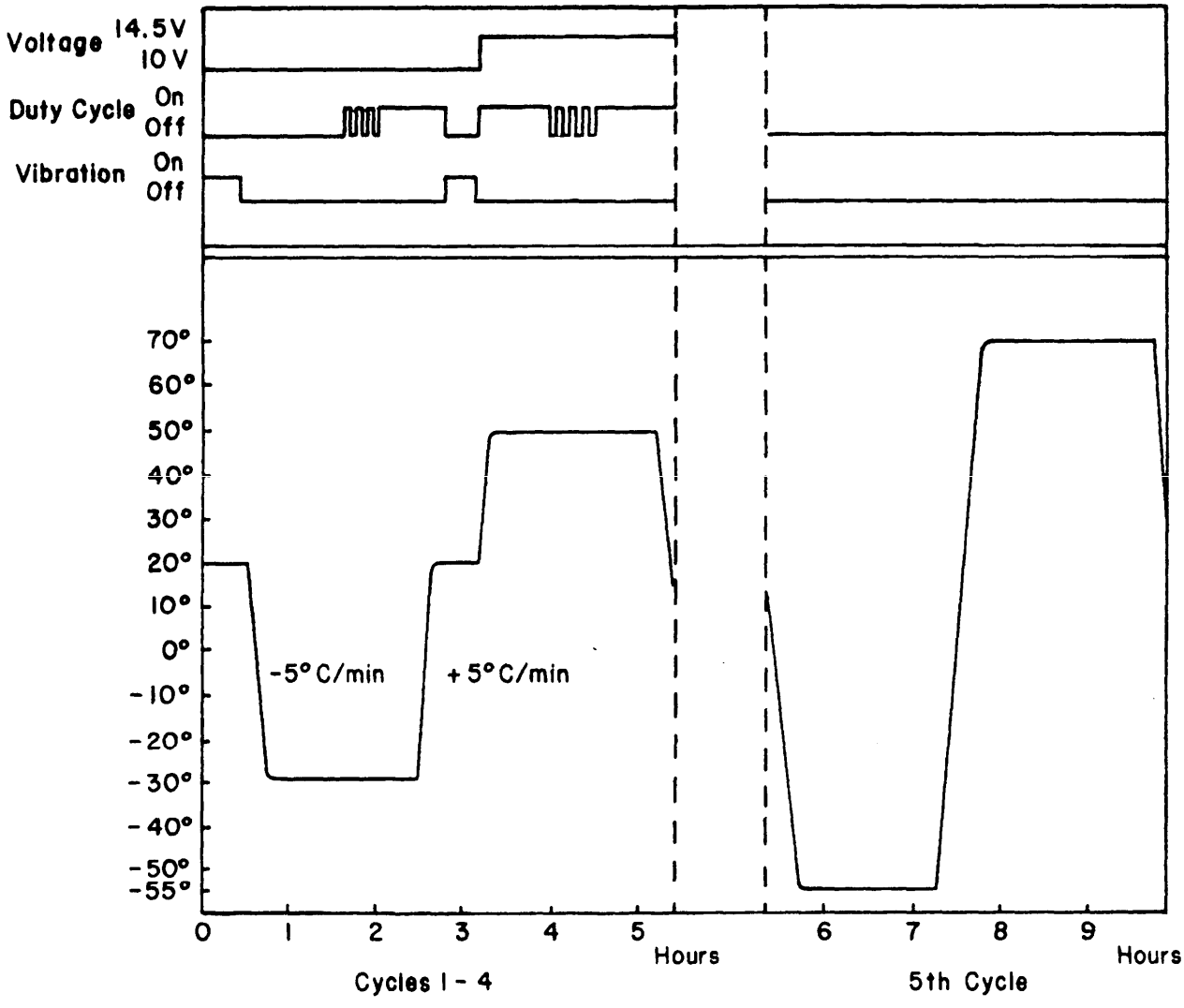
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Preparing activity:
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NOTES:

1. Power cycling shall include 3 sequences of 5 minutes on, 1 minute off, per temperature cycle.
2. Axis of each detector vibrated shall be alternated equally during the reliability testing. Each axis will be tested for 15 minutes.
3. All temperature cycles shall change at a minimum of 5° C/minute.
4. Each two hour temperature cycle may include the time necessary to reach the required temperature.

FIGURE 1. AN/PRS-7 Reliability test plan.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-52634C	Detecting Set, Mine, Portable: Metallic and Nonmetallic; AN/PRS-7

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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